



ST ANDREW'S JUNIOR COLLEGE
JC2 Preliminary Examination 2009

MARK SCHEME

Disclaimer

This mark scheme should be used purely as a revision tool.
Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Key to mark scheme

ref.	with reference to
/	alternative wordings/phrases
(brackets)	recommended wording that is optional
<u>underline</u>	accept this word only
OVP	other valid points

Paper 1

Qn No	Ans	Qn No	Ans
1	A	21	A
2	A	22	C
3	B	23	A
4	C	24	B
5	B	25	C
6	A	26	D
7	A	27	D
8	D	28	A
9	B	29	B
10	B	30	C
11	C	31	B
12	D	32	D
13	B	33	D
14	D	34	D
15	A	35	A
16	C	36	B
17	C	37	C
18	D	38	D
19	C	39	D
20	C	40	B

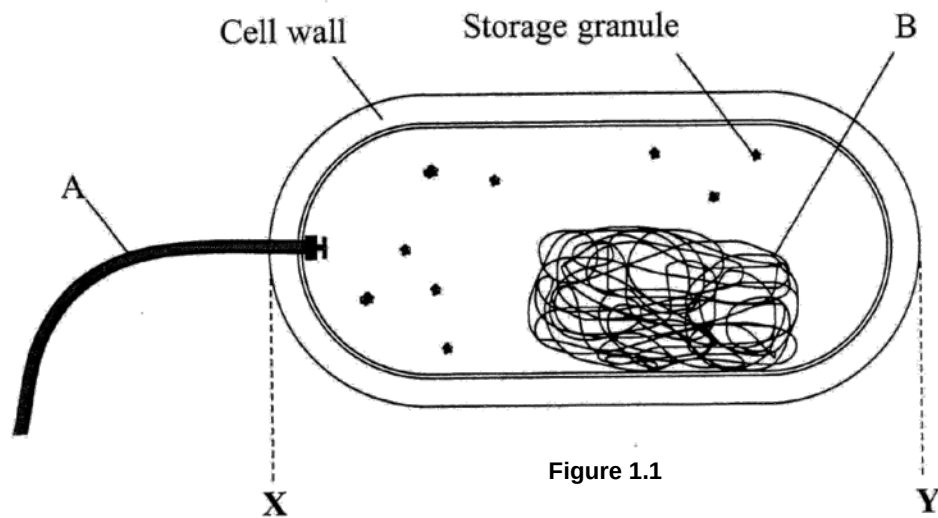
Paper 2

Section A

Answer all questions.

QUESTION 1

Figure 1.1 shows the structure of a bacterial cell as seen using an electron microscope.



(a) Name the structures labelled A and B. [1]

- 1 A = flagellum;
- 2 B = bacterial chromosome ; (reject bacterial genome)

(b) Name the carbohydrate present in the storage granules [½]

- 1 glycogen.

(c) Describe briefly how the cell wall in this bacterial cell differs from that in a plant cell. [1]

- 1 bacterial cell wall made up of peptidoglycan/ murein;
- 2 plant cell wall made up of cellulose.

- (d) The diagram has been magnified 6000 times. Calculate the actual length of bacterial cell between X and Y. Show your working and present your answer in micrometers. [1½]

- 1 Correct measurement : 86 – 87 mm;
- 2 Correct division : /6000;
- 3 Correct conversion to μm : 14.0 μm - 14.3 μm

Dental plaque is caused by the growth of bacteria on a biofilm found on teeth surface. Synthesis of the biofilm requires an enzyme to convert sucrose into dextran (an important component of biofilm).

- (e) Complete the table below to show the coding strand on DNA. [1]

Base sequence on coding strand of DNA	<u>T</u>	<u>G</u>	<u>C</u>	<u>A</u>	<u>A</u>	<u>T</u>	<u>C</u>	<u>G</u>	<u>A</u>	<u>C</u>
Base sequence on mRNA	A	C	G	U	U	A	G	C	U	G

- (f) Describe how the enzyme is assembled from free amino acids. [6]

- 1 amino acid activation / attachment of an amino acid to its specific tRNA;
- 2 aminoacyl tRNA synthetases;
- 3 binding of ribosome to mRNA (at mRNA binding site);
- 4 chain initiation / Start codon AUG positioned at P site;
- 5 binds initiator tRNA with formylmethionine;
- 6 chain elongation / amino acids are lined up in the correct sequence;
- 7 complementary base pairing via hydrogen bonds;
- 8 between anti-codon on tRNA and codon on mRNA;
- 9 formation of peptide bonds between adjacent amino acids;
- 10 catalysed by peptidyl transferase;
- 11 ribosomes translocates down mRNA by one codon / 3 nucleotides ;
- 12 chain termination / stop codon encountered;
- 13 addition of water to release polypeptide chain water;
- 14 polypeptide chain folds into its specific shape

- 15 due to hydrophobic interactions, disulphide, ionic and hydrogen bonds;**
- 16 between amino acids far away;**
- 17 2 or more polypeptide chains may interact to form a quaternary structure;**
- 18 enzyme with active site for binding to substrate is formed;**

- (g) Research is taking place to see if chemicals can be added to toothpaste to block the expression of the genes responsible for the synthesis of the sticky gel and therefore, stop plaque forming.

RNA interference is one method used to block the expression of genes. This uses RNA molecules that are complementary to the messenger RNA of the gene.

- (i) Explain how RNA interference affects the expression of a gene. [2]

- 1 prevents mRNA from combining with ribosomes ;**
- 2 stopping protein synthesis / translation ;**
- 3 ref to trigger destruction of its mRNA;**
- 4 removes transcript for translation to occur.**

- (ii) Unfortunately, adding complementary RNA to toothpaste has not proved successful in controlling plaque. Suggest 2 reasons why this is so. [1]

- 1 chemicals / enzymes in mouth / toothpaste / bacteria that denature / degrade, RNA**
- 2 RNA not normally taken up by bacterial cells ;**
- 3 Short half life of RNA ;**
- 4 RNA not replicated in bacteria when bacteria reproduce ;**
- 5 toothpaste in mouth only for a short time / easily washed away by saliva ;**

[Total : /14]

QUESTION 2

The liver plays an important role in carbohydrate metabolism. Figure 2.1 shows the principal biochemical pathways associated with glucose and glycogen in the liver of mammals.

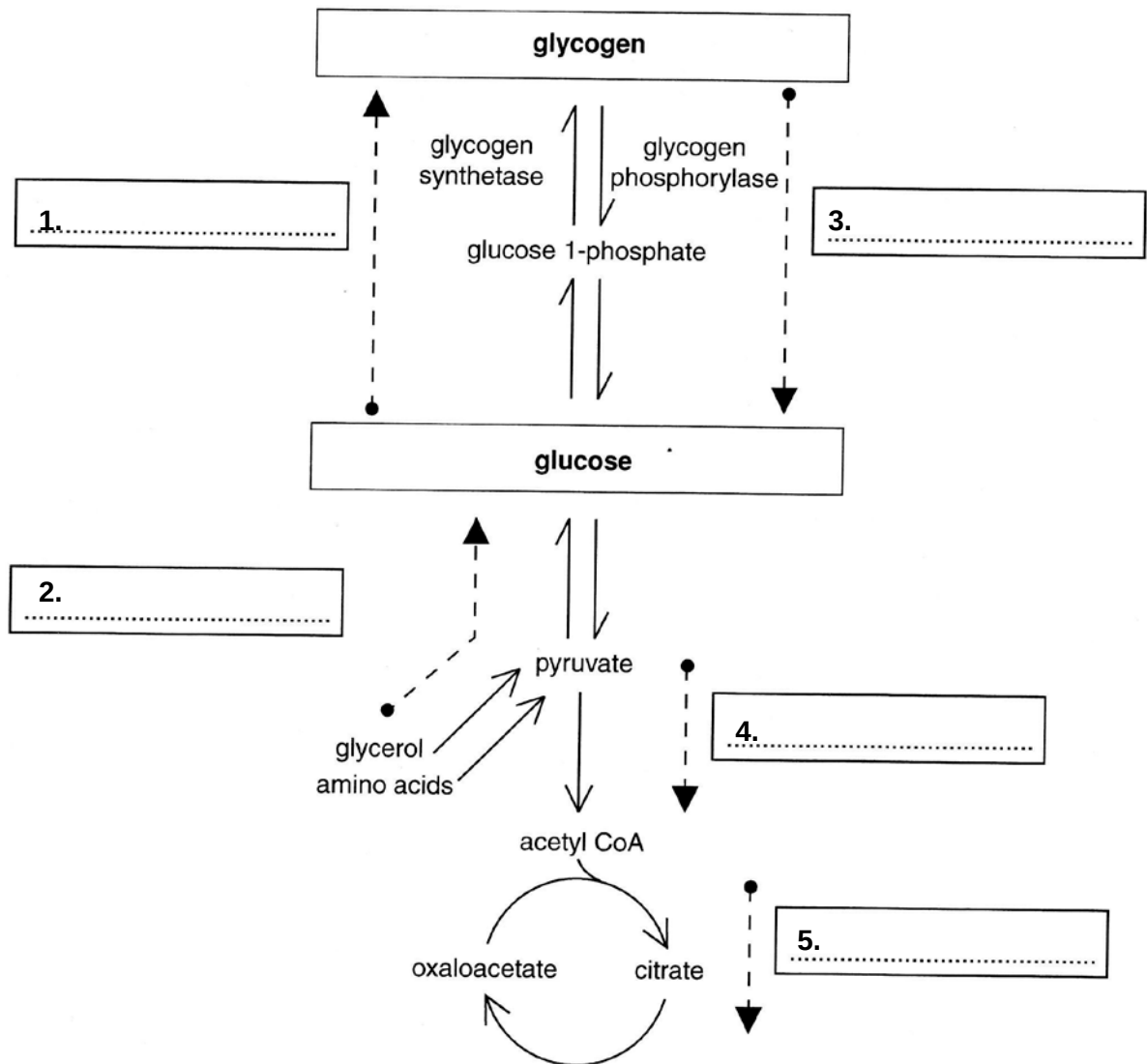


Figure 2.1

- (a) Complete Figure 2.1 by writing the names of the biochemical pathways in the spaces provided. [2½]

- 1 **glycogenesis/
synthesis of glycogen; (reject polymerization/condensation)**
- 2 **gluconeogenesis/
synthesis of glucose from non-carbohydrates derivatives;**
- 3 **glycogenolysis/
breakdown of glycogen; (reject hydrolysis)**
- 4 **link reaction;**
- 5 **Krebs cycle.**

(b) Explain why glycogen is suitable as a storage molecule in liver cells. [3]

- 1** easily hydrolysed to form monosaccharides;
- 2** due to lack of cross-linking to form big bundles;
- 3** compact shape;
- 4** large amounts of glycogen can be stored within a cell;
- 5** insoluble in water;
- 6** does not affect water potential of cells;

Figure 2.2 shows the activity of the enzymes glycogen phosphorylase and glycogen synthetase in the liver of a rat after consumption of a glucose meal.

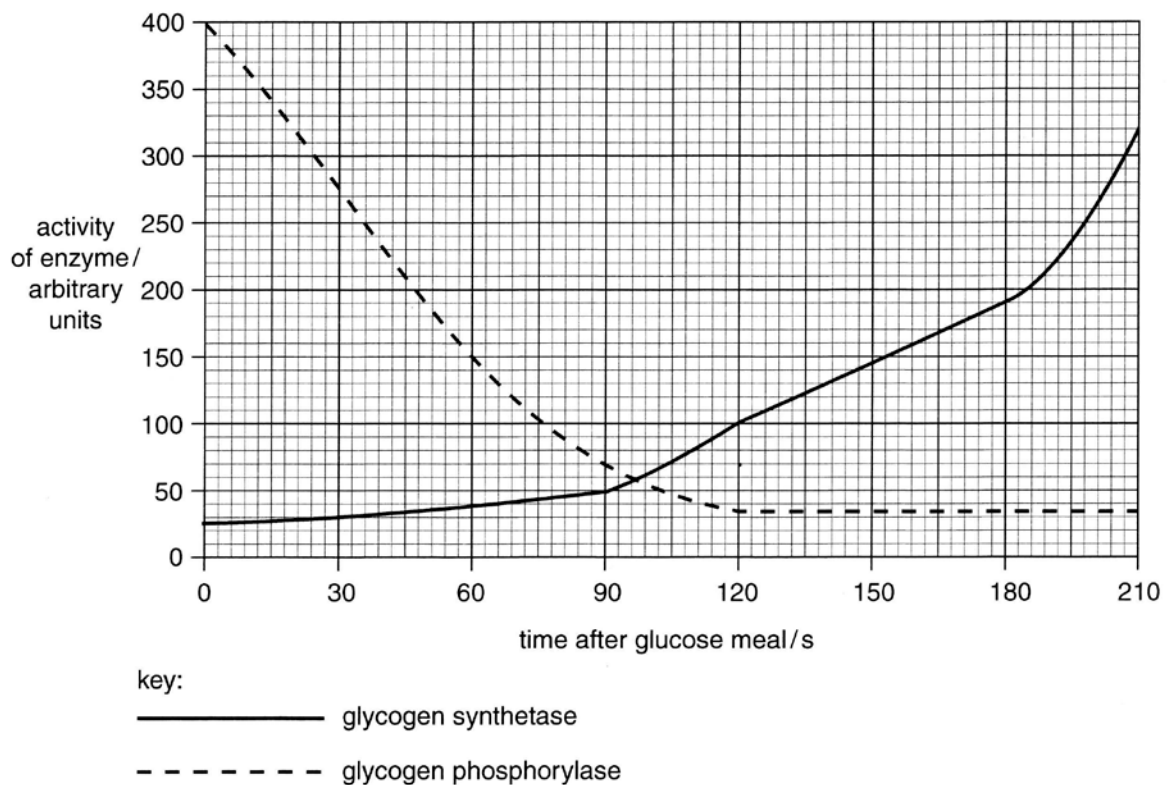


Figure 2.2

- (c) With reference to Figure 2.1, explain the changes in the activity of the two enzymes shown in Figure 2.2 following the consumption of a glucose meal. [3½]

- 1 (following glucose meal) blood glucose levels rise;
- 2 Insulin released;
- 3 (more) glucose enters the liver cells/increased substrate concentration;
- 4 results in increased activity of glycogen synthetase;
- 5 from 25 a.u at 30s to 320 a.u by 210s;
- 6 glucose converted to glycogen/glycogenesis;
- 7 results in decrease activity of glycogen phosphorylase;
- 8 ref 400 a.u at 0 min to 35 a.u by 210s;
- 9 reduced glycogenolysis/breakdown of glycogen to glucose.

- (d) The pancreas consists of both endocrine and exocrine tissue. The endocrine tissue produces the hormones insulin and glucagon, both of which are involved in the regulation of blood glucose.

- (i) Name the endocrine tissue found in the pancreas. [½]

1 islets of Langerhans

- (ii) Name the cells that secrete insulin. [½]

1 β cells;

- (e) Insulin circulates throughout the whole body. Receptors for insulin are found on cells of the liver, adipose tissue and muscle tissue.

The number of receptors on the target cells decreases if insulin levels remain high for long periods, resulting in the target cells becoming insensitive to insulin. This condition is commonly associated with obesity, where affected individuals can become insulin resistant and show the symptom of diabetes mellitus.

- (i) Suggest both the specific location and the chemical structure of the receptors. [1]

**1 location : cell surface membrane /plasma membrane;
2 chemical structure : protein/polypeptide/glycoprotein.**

- (ii) Other than the changes in enzymatic activities of glycogen synthetase and glycogen phosphorylase, state two other changes that will occur within cells possessing insulin receptors when the insulin binds to the receptors. [1]

**1 increase in membrane permeability to glucose;
2 increase in rate of conversion of glucose to fat (in adipose tissues);
3 rate of respiration increases.**

- (iii) Explain why the injection of insulin is not useful in the treatment of obesity-related diabetes. [1]

**1 insulin still being produced;
2 lack of insulin receptors;
3 cells do not respond to insulin;
4 likely to be suffering from type II/insulin-independent diabetes.**

[Total : /13]

QUESTION 3

HIV-1 protease is an enzyme produced by the human immunodeficiency virus. It is formed from two identical polypeptides, each of which is made up of 99 amino acids. The polypeptides both have a helices and b strands. The active site of the enzyme is formed by three amino acids from each polypeptide. Fig. 1.1 shows HIV-1 protease.

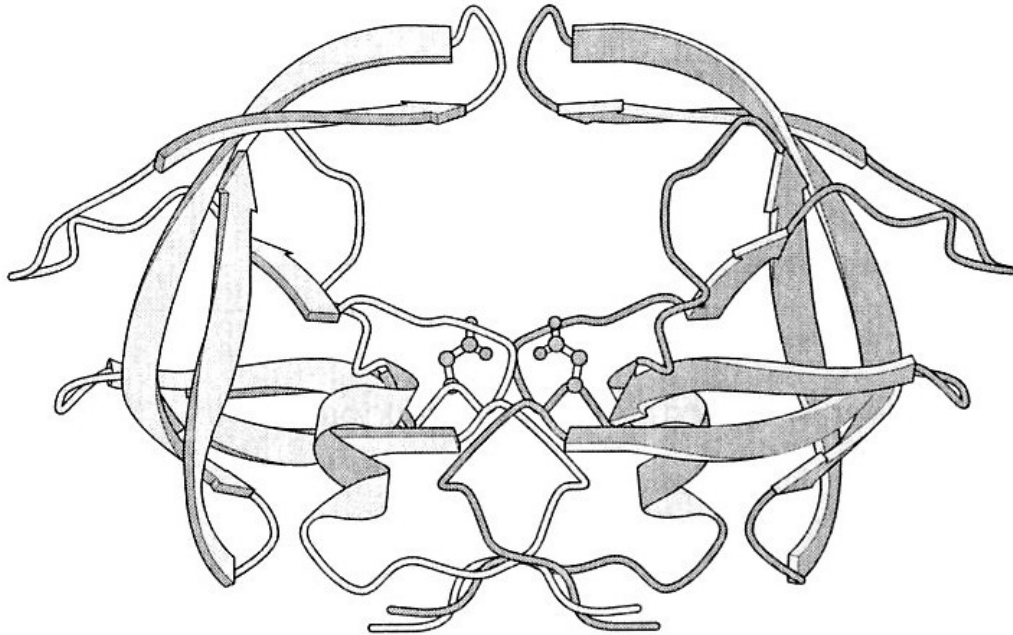


Figure 3.1

(a) With reference to HIV-1 protease, explain what is meant by primary structure? [1]

- 1 sequence of amino acids;
- 2 each polypeptide contained 99 amino acids

(b) With reference to HIV-1 protease, explain what is meant by tertiary structure? [2]

- 1 bonds between adjacent section of polypeptide chain;
- 2 e.g. ionic bonds, hydrogen bonds, disulphide bridges and hydrophobic interaction (reject van der Waals forces)
- 3 alpha helix or beta chains are folded
- 4 to form three dimensional shape
- 5 position of the three amino acids to form half the active site
- 6 globular shape

(Reject: reference to quaternary structure – not required in the question.)

- (c) Explain how polyproteins may bring about a change in the shape of the enzyme. [2]

- 1 the polyprotein (substrate) is held by H bonds and hydrophobic interactions;
- 2 with the binding residues at the active site;
- 3 conformation change induced by the binding;
- 4 results in the mobile flaps of the protease wrapping around the substrate.

- (d) Describe how HIV-1 protease brings about the splitting of polyproteins. [2]

- 1 complementary shape of substrate to the active site;
- 2 enzyme-substrate complex formed;
- 3 lowered activation energy;
- 4 split by breaking peptide bonds.

(Reject: Substrate has shape which is *similar* to the active site.)

- (e) Drugs such as Saquinavir have been developed that can stop the action of HIV-1 protease. The drug has a similar shape to regions of the polypeptide chain.

Describe how Saquinavir inhibits the action of HIV-1 protease. [2]

- 1 competitive inhibitor;
- 2 binds at the active site;
- 3 inhibitor-substrate complex;
- 4 number of enzyme-substrate complex formed reduced;
- 5 less hydrolysis of polyprotein

(Reject: Inhibitor-enzyme complex as the inhibitor has a similar shape to the enzyme.

Reject: Enzyme-substrate complex cannot be used for binding between enzyme and inhibitor.

Reject: Inhibitor binds to allosteric site / shape of active site changed / binds permanently.

Reject: Inhibitor cause inhibition of enzyme (because it is merely lifted from the question)

- (f) The viral genome is not stable and undergoes frequent mutation. Suggest why this means that researchers may need to continue to develop new drugs in order to prevent the action of HIV-1 protease. [2]

- 1 Mutations in the viral genome will alter the genes ;
- 2 Encoding for Either the polypeptide (substrate) and/or HIV-1 protease (enzyme);
- 3 Change in the shape of the protein
- 4 Hence inhibitor may no longer be complementary to the active site

(Reject: Mutation in the genome resulted in a change to the active site of the genome.)

Many drugs have been designed to make it possible for AIDS patients to improve their overall health and extend their lives. The emergence of a new family of drugs known as Azidothymidine (AZT), has been shown to affect the reproductive cycle of HIV. The structures of AZT and thymidine are illustrated in **Figure 1.2** below.

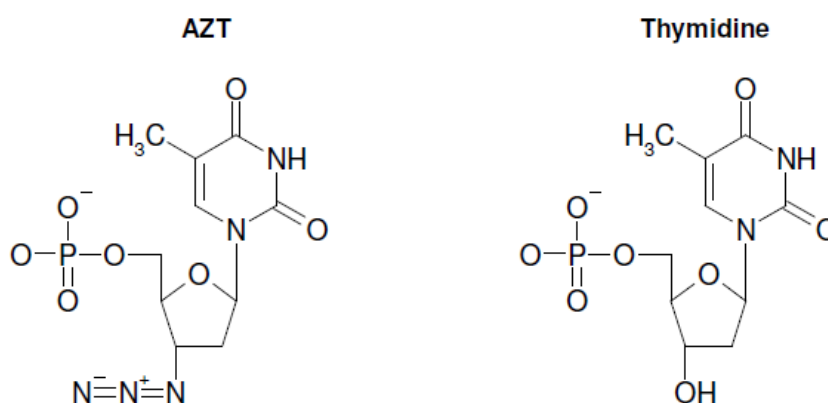


Figure 3.2

- (g) Explain how AZT affects the reproductive cycle of HIV. [2]

- 1 AZT affects the replication / DNA synthesis stage of the HIV replication cycle;
- 2 AZT is structurally similar to thymidine but lacks free 3'OH group (replaced by $\text{N}^-=\text{N}^+=\text{N}$ group)
- 3 Disallows the further addition of nucleotides / prevent the formation of phosphodiester bonds
- 4 Facilitates the DNA chain termination / blocks further chain elongation

(Reject: AZT added into RNA – as uracil is used in place of thymidine in RNA)

[Total : /13]

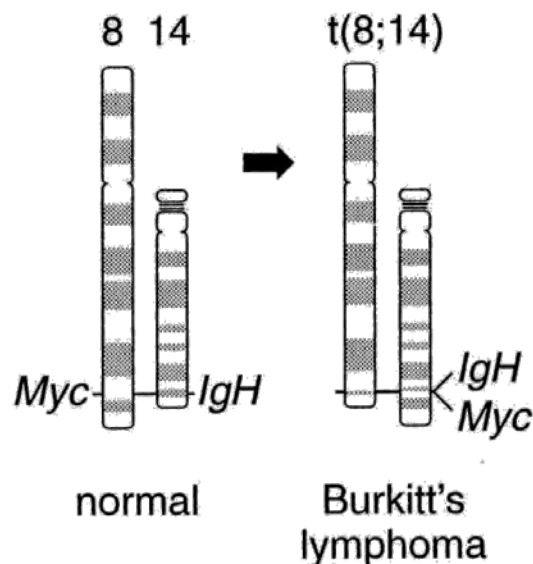
QUESTION 4

(a) Explain how one named factor can increase the chances of a cancerous growth. [3]

- 1 ionizing radiation e.g. UV light, gamma rays, X-rays/
chemical carcinogens e.g. tar in cigarette smoke/
viruses e.g. human papillomavirus;
- 2 may induce mutations in DNA;
- 3 ref proto-oncogene;
- 4 leads to activation of oncogene
- 5 resulting in increased cell division (reject cell growth) /gain-of-function mutation;
- 6 ref tumour suppressor genes;
- 7 inactivation of both tumour suppressor genes
- 8 results in inability to slow down cell cycle/loss-of-function mutation.

- The Myc protein acts as a signal for cell proliferation.
- The gene for Myc is a proto-oncogene and it is found on chromosome 8.
- In Burkitt's lymphoma, a cancer involving B lymphocytes, a chromosome translocation shifts the Myc gene to chromosome 14.
- This brings Myc gene close to powerful gene regulatory sequences.
- These regulatory sequences normally control the production of large amounts of antibodies (IgH) from each cell.
- In the mutant cells, however, they cause excessive expression of the Myc gene, and this causes the mutant B lymphocytes to proliferate and form a tumour.

Figure 4.1 shows this chromosome translocation.



(b) State what you understand by the terms

(i) Proto-oncogene [2]

- 1 involved in normal cell division;**
- 2 encodes proteins involved as transcription factors stimulating expression of other genes;**
- 3 encodes signal transduction molecules that stimulate cell division;**
- 4 encodes cell cycle regulators.**

(ii) oncogene. [1½]

- 1 results in cancer;**
- 2 mutated form of proto-oncogene;**
- 3 results in uncontrolled cell division;**

(c) Explain how the Myc proto-oncogene becomes an oncogene in Burkitt's lymphoma. [2]

- 1 translocation of Myc proto-oncogene near to a gene regulatory sequence;**
- 2 ref highly active promoter**
- 3 ref increased transcription of Myc gene;**
- 4 increased translation of Myc mRNA;**

- (d) During the development of cancer, tumour cells that are exposed to one anticancer drug often develop a resistance not only to that drug but also to other drugs to which they have never been exposed.

Multiple-drug resistance is frequently linked with amplification of a part of the genome that contains a gene called Mdr1.

The Mdr1 gene codes for a membrane-bound transporter.

Suggest how amplification of Mdr1 will allow tumour cells to show multi-drug resistance. [2]

- 1 gene amplification results in more copies of Mdr1;**
- 2 leading to more copies of the transporter;**
- 3 drugs would be removed from the cell;**
- 4 causing the resistance to drugs;**

- (e) Suggest how this multi-drug resistance in a tumour may be overcome. [½]

- 1 use of an inhibitor to block transporter;**

[Total : /11]

QUESTION 5

A pure-breeding variety of tomato plant, variety A, produced red fruit which remained green at their bases even when ripe.

Plants of variety A were crossed with another pure-breeding variety, B, with orange fruit which have no green bases when ripe. The F_1 generation plants all had red fruit with green bases.

- (a) Using the symbols G/g to represent the alleles at the locus controlling green-based and non green-based fruits and the symbols R/r to represent the alleles at the locus controlling red or orange fruit, state the genotype of variety B. [½]

1. **rrgg**

- (b) Plants from the F_1 generation were test crossed to variety B. Using the symbols given in (a), draw a genetic diagram of the test cross to show the expected ratio of offspring phenotypes. [4]

Parental phenotypes:	Red fruit Green bases	x	Orange fruit No green bases
Parental genotype:	RrGg	x	rrgg
Parental gametes:	RG Rg rG rg	x	rg

Punette square showing offspring genotypes (optional)

	RG	Rg	rG	rg
rg	RrGg	Rrgg	rrGg	rrgg

Offspring genotype	RrGg	Rrgg	rrGg	rrgg
Offspring phenotype	Red fruit Green base	Red fruit Non-green base	Orange fruit Green base	Orange fruit Non-green base
Offspring ratio	1	1	1	1

1. **parental genotype – RrGg;**
2. **parental gametes – RG, Rg, rG, rg;**
3. **parental gametes – rg;**

4. **Offspring genotypes corresponds to phenotypes; [1m]**
5. **Ratio of 1:1:1:1; [1m]**
6. **Circled gametes.**

(c) Two hundred randomly chosen offspring from the test cross described in (c) had the following phenotypes :

Red fruit with green bases	55
Red fruit with no green bases	45
Orange fruit with green bases	43
Orange fruit with no green bases	57

The χ^2 (chi-squared) test was performed on these data, giving a calculated value for χ^2 at 3.2.

(i) State the number of degrees of freedom applicable to these data. [½]

1. 3.

Degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

(ii) Use the calculated value of χ^2 and the table of probabilities provided in Table 5.1 to find the probability of the results of the test cross departing significantly by chance from the expected ratio. [½]

1. **probability greater than 0.10/
probability>0.1;
probability>10%.**

- (iii) state what conclusions may be drawn from the probability found in (d)(iii) [1½]

1. probability is above 0.05;
2. difference from expected results is not significant/statistically different;

and

3. observed result is due to chance;

or

4. null hypothesis should be accepted/alternative hypothesis rejected;

or

5. loci for 2 genes assort independently/not linked;

- (d) Experiments have shown that loci G/g and R/r are on the same chromosome of the tomato plant genome. The two loci are 44 map units apart. Explain how the results of the test cross shown in (c) could occur when the two loci are on the same chromosome. [2]

1. ref partial linkage;
2. map units indicate loci far apart (for linkage to be detected);
3. recombinants of phenotypes seen;
4. some detail of crossing over/
e.g. ref prophase I of meiosis/exchange of non-sister chromatid;
5. crossing over has occurred twice/even number of times;
6. restoring loci to parental combinations.

[Total: /9]

QUESTION 6

Fig. 6.1 shows dendrites from each of two neurones, **A** and **B**, forming synapses with a third neurone, **C**. Concentrations of some ions are also shown.

	Ion concentration / nmol dm^{-3}		
	Na^+	K^+	Cl^-
Outside neurone C	145	12	120
Inside neurone C	12	155	5

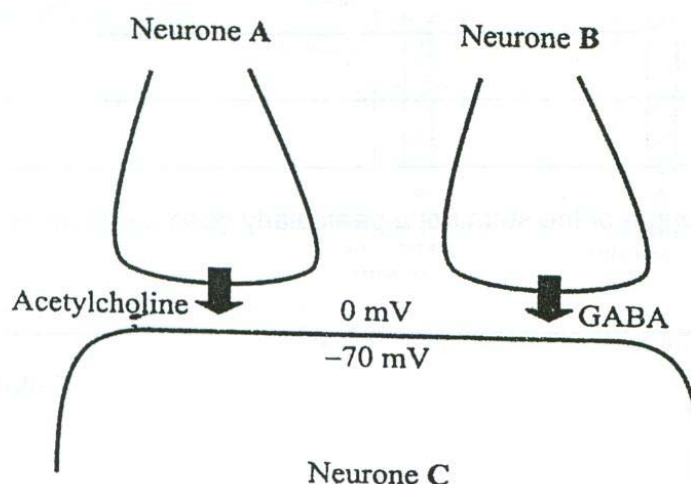


Fig 6.1

- (a) Acetylcholine increases the permeability of the postsynaptic membrane to Na^+ ions. Explain how acetylcholine released from neurone **A** generates an action potential in neurone **C**. [2]

- 1 Acetylcholine diffuses across the synaptic cleft;
- 2 binds to receptors/chemical-gated Na^+ channels on neurone **C**/ post-synaptic membrane;
- 3 triggers opening of chemical-gated Na^+ channels, causing Na^+ influx;
- 4 If post-synaptic membrane is depolarized till threshold potential action potential is generated;
- 5 ref temporal summation.

- (b) Neurone **B** releases a substance called GABA as its neurotransmitter. GABA increases the permeability of the postsynaptic membrane to Cl^- ions. Suggest how impulses in neurone **B** would reduce the chances of an action potential in neurone **C**. [2]

- 1 Chloride ions enter neurone C;**
- 2 membrane more negative than resting potential / membrane is hyperpolarized;**
- 3 threshold potential is not reached and action potential not generated in postsynaptic membrane;**
- 4 more stimulation / impulses / high acetylcholine level from A to generate an action potential;**

- (c) Explain the benefits of neurone **C** synapsing with more than one neurone. [2]

- 1 allow for integration of stimuli from various sources;**
- 2 to produce a coordinated response;**
- 3 ref. to spatial summation;**
- 4 impulse transmission can be better controlled and fine tuned to suit circumstances.**

- (d) The acetylcholine receptor is a type of G-protein-linked receptor. When acetylcholine binds to it, a membrane-bound enzyme is activated. Name the enzyme. [1]

- 1 Adenylyl cyclase / adenylate cyclase / adenylyl cyclase.**

- (e) Explain the role of the enzyme named in (d) in signal transduction that leads to a response in the postsynaptic neurone. [2]

- 1 Activated adenylyl cyclase converts ATP to cAMP;**
- 2 cAMP acts as a secondary messenger;**
- 3 activates a protein kinase (A) / cAMP dependent protein kinase;**
(whose activity is dependent on the level of cyclic AMP (cAMP) in the cell)
- 4 activates / phosphorylates other protein kinases/relay proteins/ causes phosphorylation cascade.**

[Total: 9]

QUESTION 7

The New Zealand Robin or the native bush Robin, *Petroica australis*, is a sparrow-sized bird found only in New Zealand, where it has the status of a protected endemic species. The birds are sparsely distributed through South Island and Stewart Island/Rakiura, although the distribution is not continuous. There are two species and three sub-species of these robins living in New Zealand today.



Figure 7.1

- (a) Complete Table 7.1 to show the classification of one of the New Zealand bush robin subspecies. [3½]

Kingdom	Animalia
Phylum	Chordata
Class	Aves
Order	Passeriformes
Family	Petroicidae
Genus	<i>Petroica</i> (to be underlined)
Species	<i>Petroica australis</i>
Subspecies	<i>Petroica australis australis</i>

Table 7.1

It is believed that the robins evolved from a common ancestral stock, members of which flew from Australia across the Tasman Sea and became established in New Zealand over a million years ago.

This ancestral form is considered to be similar to the present day Australian flame robin, *Petroica multicolor*, a bird with a brightly coloured red breast. The New Zealand birds do not have this red colour.

Some characteristics of the birds and their distributions are shown below:

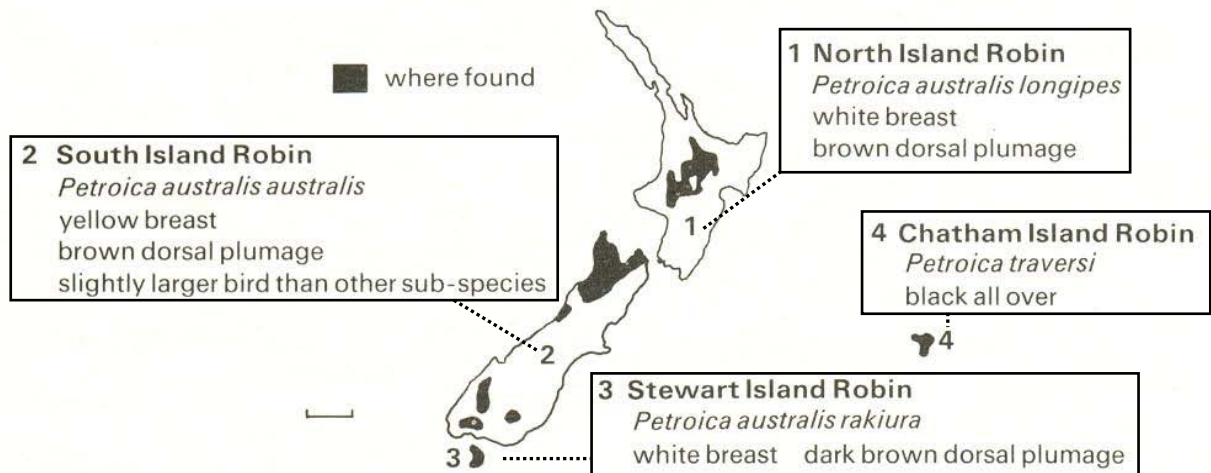


Figure 7.2

(b) Explain briefly what is meant by the term 'species'. [2]

- 1 organisms with similar physiological, morphological, behavioural, biochemical features;
- 2 capable of interbreeding;
- 3 ref viable and fertile offspring;
- 4 reproductively isolated from other species.

(c) Explain why the robins in locations 1, 2 and 3 are similar but different from those in location 4. [2½]

- 1 birds in locations 1, 2 and 3 are capable of interbreeding;
- 2 ref short distance between the islands;
- 3 subjected to similar selection pressure/
gene flow is not interrupted/
shared a common gene pool for a long time;
- 4 birds on location 4 could not interbreed with other birds;
- 5 geographically isolated ;
- 6 ref longer distance between the islands;
- 7 mutation such as the gene for black plumage would tend to be confined within the isolated inbreeding population on island 4.

- (d) Describe how the molecular clock can be used to date evolutionary events, such as when two species of native bush robins in New Zealand diverged. [2]

- 1 Based on the assumptions that neutral mutations are fixed at a relatively constant rate and used as molecular clock;
- 2 compare DNA sequence of a particular common gene (between the two native bush robin species);
- 3 determine rate of evolutionary change in another gene / DNA sequence;
- 4 number of mutation / substitution / difference in genetic sequence is used to calculate the length of time since divergence.

- (e) The red breast trait is present in Australian robins. Account for the absence of this trait in New Zealand robins. [1]

- 1 Australian robins might have gained the red breast by a mutation (which occurred in the Australian population)
- 2 after the colonization of New Zealand by forms which did not possess it/ref to Founder effect;

OR

- 3 red-breasted form might have reached New Zealand;

and

- 4 characteristic was at a selective disadvantage/
ref genetic drift e.g. bottleneck events/
red breast individuals accidentally die;
- 5 frequency of red breast allele decreased and eventually disappeared from the gene pool.

[Total : /11]

Section B

QUESTION 8

- (a) Describe how a bacteriophage can infect a bacterium, eventually forming an infected bacterial colony. [8]

1. lysogenic cycle of λ bacteriophage;
2. phage recognizes surface receptors on host's cell membrane;
3. binds to surface of host cell;
4. injection of DNA into the bacterial cytoplasm;
5. incorporation of viral genome into host genome;
6. at specific insertion sites;
7. viral DNA known as prophage;
8. transcription of most prophage genes halted
/ phage genome mostly silent within bacterium;
9. DNA of bacterial chromosome replicates at origin of replication;
10. via semi-conservative replication;
11. involving DNA polymerase;
12. 2 ori move to each opposite end of cell;
13. cell elongates and plasma membrane grows inwards;
14. plasma membrane grows inwards;
15. division of parent bacterium into 2 daughter cells;
16. each daughter cell inherits a complete genome;
17. binary fission continue to occur until a colony is formed;
18. viral genome replicates along with host genome;
19. copies of phage genome pass on to daughter cells;
20. large population of bacteria carrying virus in prophage form;

- (b) Using *lac operon* as an example, explain how prokaryotes can regulate their gene expression in response to changes in concentration of glucose and lactose in their growth medium. [8]



1. Lac Z, Lac Y and Lac A;
2. codes for β -galactosidase, permease and transacetylase;

When [lactose] is low or absent

3. repressor coded for by regulatory gene (Lac I);
4. attachment of repressor to operator;
5. inhibits the attachment of RNA polymerase to promoter;
6. no transcription of 3 structural genes;

When [lactose] is high

7. allolactose binds to repressor;
8. changes conformation of repressor;
9. repressor can no longer bind to the operator;
10. RNA polymerase can now bind to promoter;
11. transcription of genes can occur;
12. negative control – genes are not transcribed when repressor is bound;

When [glucose] is low or absent

13. high adenyl cyclase activity, high [cAMP];
14. cAMP binds to CAP (catabolite activator protein);
/ CRP to form cAMP-CAP complex;
15. binds to CAP binding site;
16. enhances the attachment of RNA polymerase;
17. increase transcription;
18. positive control – increase in the production of enzymes;

When [glucose] is high

19. low adenyl cyclase activity, low [cAMP];
20. no binding of cAMP-CAP complex to CAP binding site
/ no cAMP-CAP complex is formed;
21. binding of RNA polymerase is not enhanced;
22. no / low transcription of structural genes;
23. even if lactose is present as glucose is the preferred substrate;
24. known as catabolite repression/ glucose repression;

- (c) State the main differences between the structure and organization of prokaryotic and eukaryotic genome. [4]

Features	Prokaryotic Genome	Eukaryotic Genome
Location	Within a region called the nucleoid in within the cytoplasm	Within membrane bound nucleus
Chromosome no.	Single chromosome (haploid)	Several paired chromosomes
Chromosome structure	Circular chromosome, double helix forms a covalently closed circle	Linear chromosome with two ends
Packing of DNA	DNA is not packaged into nucleosomes	DNA is packed into nucleosomes
Amount of DNA	Less total DNA per cell than eukaryotes	Usually more total DNA per cell than prokaryotes
Gene density	More compact genetic organisation	Presence of more intergenic spaces
Organisation of genes	Most of the DNA codes for protein, tRNA or rRNA Small amount of non-coding DNA mainly consisting of regulatory sequences	Most of the DNA does not encode protein or RNA More complex regulatory sequences
Presence of introns	No introns, no discontinuous genes in prokaryotic genomes	Introns are present interrupting the coding sequence, exons
Cluster organization of genes	Two or more genes may be expressed and regulated together as a unit Genome arranged in operons	Genes may not be regulated together Absence of operon
Extra-chromosomal genes	May contain plasmids	Usually do not contain plasmid / presence of mitochondrial / chloroplast genome
Location	Within a region called the nucleoid in within the cytoplasm	Within membrane bound nucleus

QUESTION 9

(a) Discuss the roles of Natural selection and genetic drift in evolution. [10]

1. Evolution is the accumulated changes to the frequencies of many alleles in the gene pool of a population over a period of time;

Natural Selection

2. Operates on populations in nature;
3. More pronounced in small populations;
4. Populations tend to over produce / have great reproduction potential;
5. But populations actually maintain relatively constant number;
6. Due to environmental factors being selective agent or pressure;
7. Selection processes include directional selection, stabilizing selection and disruptive selection;
8. Variation among offspring;
9. Selecting for organisms with favorable combination of characteristics;
10. Determined by genes;
11. To survive longer, reproduce and pass on these favorable traits to their offspring;
12. Allele frequencies of such traits will increase in the population / gene pool;
13. Selecting against organism with unfavorable traits / allele frequencies of undesired traits will decrease in population or gene pool;

[max 5]

Genetic drift

14. Major cause of evolution in small populations;
15. As the relatively small numbers of chance variations in gene frequencies in large populations are absorbed into the population in succeeding generations and negligible overall effect;
16. Causes chance fluctuations in gene pool frequencies from one generation to the next / can result in either the fixation of that allele or its elimination from the gene pool;
17. By unpredictable / chance / random events;
18. Genetic drift interacts with the processes by which small populations become established to produce major evolutionary changes, i.e. bottleneck effect and founder effect;

Bottleneck effect

19. Occur when a population's size is reduced for at least one generation;
20. Undergoing a bottleneck can reduce a population's genetic variation by a lot;
21. Population may not be able to adapt to new selection pressures;
22. As the genetic variation that selection would act on may be out of the population;
23. E.g. northern and southern elephant seals
24. Northern elephant seals have much less genetic variation than southern elephant seals due to intense hunting which reduced their population size to being very few;

Founder effect

25. Occur when a new colony is started by a few members of the original population;
26. Colony may have reduced genetic variation from the original population;
27. A non-random sample of the genes in the original population;
28. E.g. the Afrikaner population of Dutch settlers in South Africa
29. Unusually high frequency of the gene that causes Huntington's disease as those original Dutch colonists just happened to carry that gene with unusually high frequency;

[max7]

- (b) Discuss the roles of biogeography and fossil records in supporting evolution.
[6]

Biogeography

1. The study of the geographic distribution of species;
2. Reflect evolutionary history as well as interactions with both living and non-living components of the environment;
3. Closely related species tend to be found in the same geographic region;
4. E.g. island biogeography
5. Most island species are closely related to species from the nearest mainland or neighboring island;
6. Same ecological niches in distant regions are occupied by different but similar looking species;
7. E.g. sugar gliders and “flying” squirrels
8. While the marsupial sugar gliders superficially resemble the eutherian “flying” squirrels, the ability to glide through the air evolved independently in these two distantly related groups of mammals;

[max 3]

Fossil records

9. Impressions of past organisms preserved in rock;
10. Parts of the organism, such as bones, shells and teeth, rich in minerals harden as fossils;
11. Used to draw phylogenetic trees;
12. Support the idea that diversity of life arose from evolution;
13. Deeper layer contains fossils of older organisms;
14. Discovery of transitional fossils link older organisms to modern species;
15. Show new species appearing and others disappearing;
16. May be due to mass extinction of species living at the time and new species repopulating the area through migration;

[max 3]

- (c) Assess the reliability between anatomical and molecular data in tracing evolution. [4]

Anatomical characteristics

1. Analogous structures / analogous adaptations / homoplasies may have similar functions;
2. But no common ancestry;
3. E.g. wings of insects and wings of birds / fins of fishes and flippers of dolphins;
4. Homologous structures have different functions;
5. Resulting from a shared ancestry;
6. E.g. flipper of dolphin and forelimb of human, based on the same five-digit plan;
7. Homologous structures more reliable than analogous structures in tracing evolution;

[max 2]

Molecular data

1. Includes DNA and amino acid sequences;
2. are unambiguous / objective;
3. Offers limitless set of characters;
4. As organisms have millions to billions of nucleotide positions;
5. Can be used to examine all kinds of taxa;
6. As all known life based on nucleic acids;
7. Quantitative and convertible to numerical form for mathematical and statistical analysis creating distance matrix and similarity matrix;
8. Molecular data more reliable than anatomical data in tracing evolution;

[max 2]

Paper 3

Section A

Answer all questions.

QUESTION 1

Figure 1.1 shows an insulin molecule. It is composed of two polypeptide chains : an A chain of 21 amino acids and a B chain of 30 amino acids. These two chains are held together by disulphide bonds formed between two amino acids.

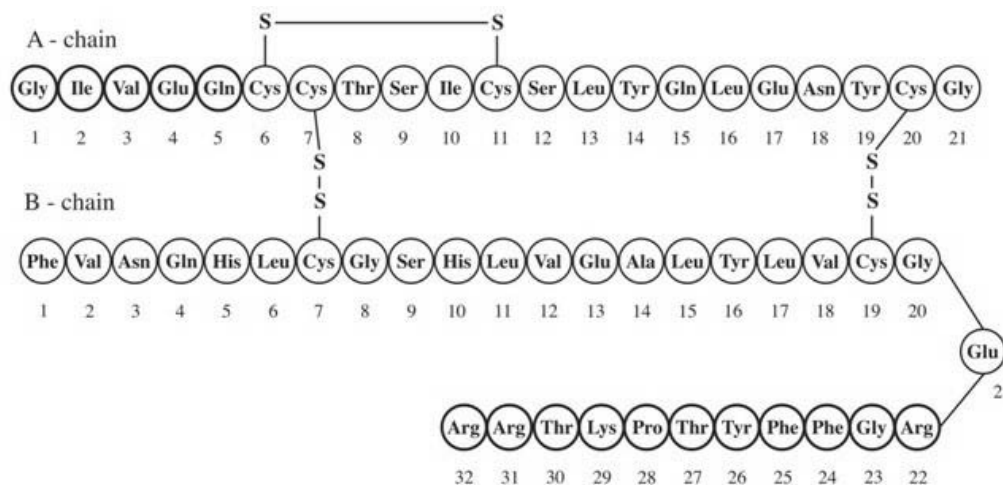


Figure 1.1

Currently, the production of recombinant insulin involves artificially synthesising the genes for the A chain and the B chain and then inserting each into 2 separate plasmids such that two recombinant plasmids are constructed (Figure 1.2), one carrying the gene for the A chain and one the gene for the B chain.

The A chains and B chains are then synthesised separately in two different mixtures of *E.coli*. The two protein chains are then extracted and purified. Finally, the A and B protein chains are mixed under special conditions allowing disulphide bonds to form *in vitro*.

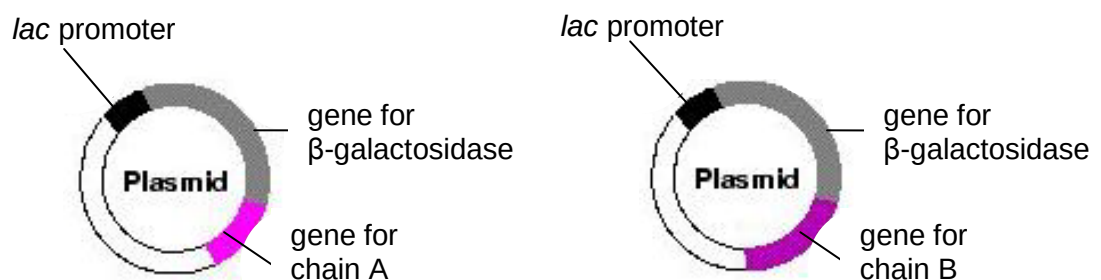


Figure 1.2

(a) State the function of the β -galactosidase gene in the plasmids. [1]

- 1 functions as a selectable/genetic marker;
- 2 to select for transformed cells with plasmid.

(b) Why is the *lac* promoter added in both recombinant vectors? [2]

- 1 ref expression vectors;
- 2 prokaryotic promoter;
- 3 allows recognition by host cell's RNA polymerase (which is also prokaryotic);
- 4 transcription initiation;
- 5 enables efficient gene expression.

(c) Explain how cells must be treated for transformation to occur. [2]

- 1 bacterial cells must be made competent;
- 2 by the addition of calcium ions (in the form of CaCl_2 solution);
- 3 bacterial cells subjected to brief heat shock treatment/electroporation;
- 4 to create transient pores in the bacterial cell membrane to allow uptake of naked DNA from the medium.

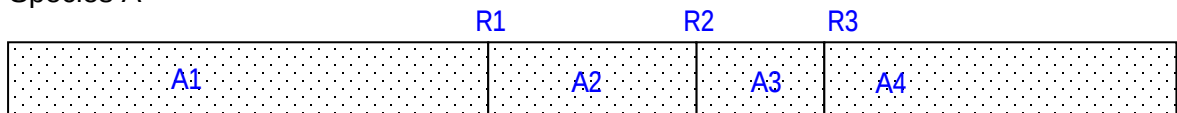
(d) The recombinant chain A and B protein produced as described above is identical to naturally occurring human insulin chains A and B. However, the synthetic gene inserted into the plasmid may possess a different nucleotide sequence to the actual human gene. Give reasons why this is so. [1]

- 1 some amino acids are encoded by more than one codon;
- 2 different nucleotide sequences in gene may still result in the same amino acids being encoded for;
- 3 reference to introns sequences in human gene;
- 4 prokaryotes do not have spliceosomes and hence introns removed from genes used for cloning.

Restriction enzymes can be used to cut DNA at specific sites called restriction sites. Genes can be cut from the chromosome of one species and joined to the chromosome of another species forming recombinant (hybrid) DNA.

Figure 1.3 shows two chromosomes from different species. The specific restriction sites (**R1**, **R2** and **R3**) and the fragments resulting from the restriction digests (**A1**, **A2**, **A3**, **A4**, **B1**, **B2**, **B3** & **B4**) are shown.

Species A



Species B

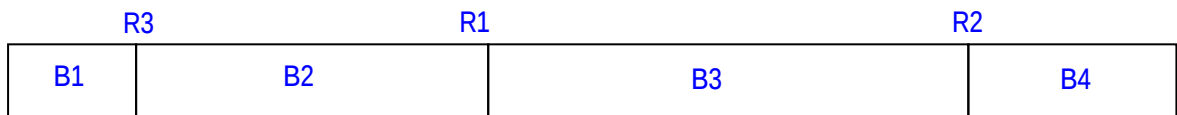
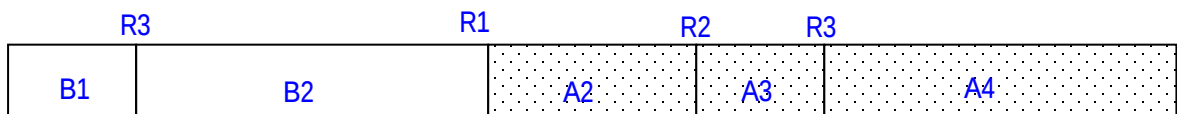
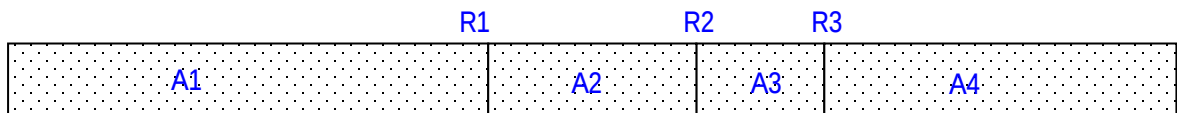


Figure 1.3

Both chromosomes were cut at restriction site R1. The resulting fragments with sticky ends were mixed and allowed to join to form recombinant (hybrid) DNA.

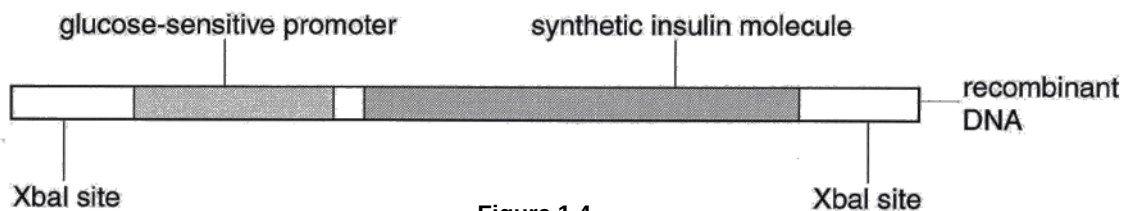
- (e) In the space below, make labelled sketches of all the possible ways in which the fragments could join. Label all fragments clearly. [3]



- 1 1 mark for 2 originals;
- 2 1 mark for each of the 2 recombinants.

Using genetically engineered bacteria, a synthetic insulin molecule has been produced which has over 20% of the activity of naturally producing insulin. The gene for the molecule has been used in an experimental gene therapy for insulin-dependent diabetes.

DNA coding for the synthetic molecule was combined, as shown in Figure 1.2, with a promoter region taken from a gene for an enzyme found only in liver cells. The promoter is sensitive to blood glucose concentration and switches on its associated gene as blood glucose concentration rises above the normal concentration.



- (f) Explain the advantage of being able to produce the synthetic insulin molecule in response to a rise in blood glucose concentration. [2]

- 1 ref. homeostasis control of blood glucose levels/
negative feedback of glucose concentration in blood;
allows for self regulation of blood glucose levels;
- 2 synthetic insulin only made when blood glucose is high;
- 3 to reduce blood glucose concentration;
- 4 restores normal set point;
- 5 no injection required for diabetics/produced on site.

Four weeks after receiving this gene therapy, the experimental diabetic rats were given a glucose meal. The concentrations of blood glucose and synthetic insulin in their blood were measured at intervals for eight hours. A control group of non-diabetic rats, which had not received any gene therapy, were given a glucose meal and the concentrations of blood glucose and insulin measured at the same time intervals. The results are shown Figure 1.5.

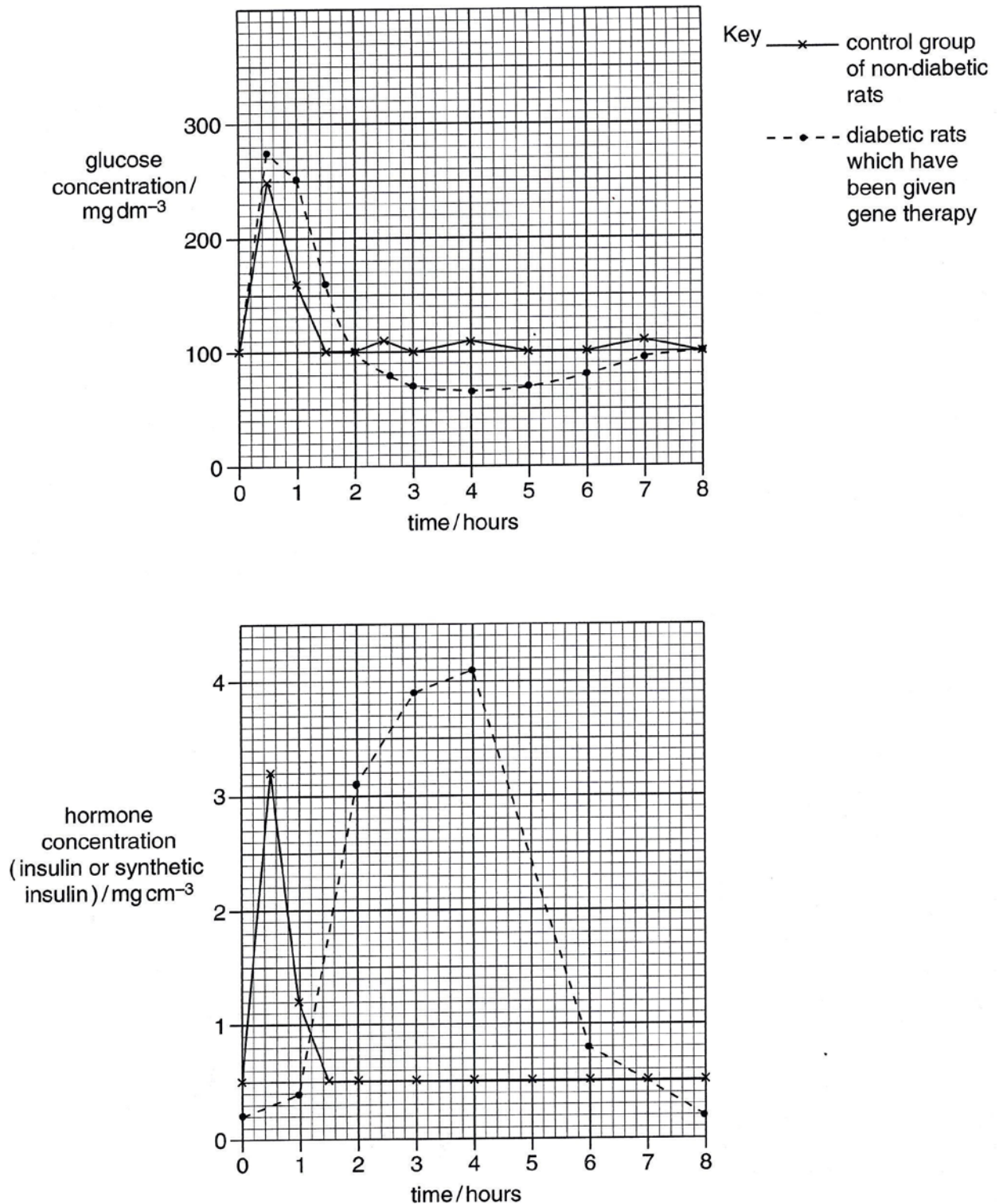


Figure 1.5

With reference to Figure 1.5, compare the concentrations of glucose and hormone in the blood of experimental and control rats. [4]

Glucose

- 1 similar pattern in both graphs;
- 2 peak slightly higher in experimental rats;
- 3 ref 275mg/dm^3 and 250mg/dm^3
- 4 drop after peak slightly delayed in experimental rats;
- 5 experimental rats had lower blood glucose levels from 2-7h;
- 6 blood glucose levels returned to normal at 7-8hr;

Hormone

- 7 peak in experimental rat higher than normal rats;
- 8 ref 3.2mg/dm^3 and 4.1mg/dm^3
- 9 peak occurs much later;
- 10 4 hr v. 30 min;
- 11 higher concentration lasts much longer;
- 12 5(6) hr v. 90 min;

[Total : /15]

QUESTION 2

Babies born with severe combined immune deficiency (SCID) have no defence against common infections and quickly become ill when the protection from maternal antibodies is lost. This was the result of a mutation in their ADA gene that resulted in infants suffering from a deficiency in all functions of their immune system.

Figure 2.1 shows the pedigree which traces the inheritance of SCID in a family.

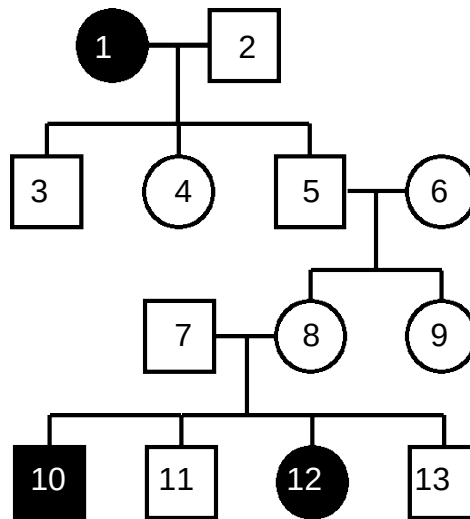


Figure 2.1

(a) Comment on the mode of inheritance of ADA-deficiency SCID. [2]

- 1 **autosomal;**
- 2 **recessive;**
- 3 **sufferers must be homozygotes/have two mutated alleles;**
- 4 **meaning both parents (if carriers) must pass on the defective gene for any of their children to get the disease /ref to individuals 7 and 8;**
- 5 **heterozygotes are carriers as they carry only one copy of the faulty gene ;**
- 6 **the chance of 2 carriers having an affected child is 1 in 4.**

(b) Explain the genetic basis of ADA-deficiency and explain how it could lead to SCID. [3]

- 1 mutation on chromosome 20;
- 2 ref non-functional enzyme;
- 3 ref adenosine deaminase;
- 4 involved in purine metabolism
/ breaking down adenosine & deoxyadenosine;
- 5 **toxic by products (deoxyATP) accumulates;**
- 6 harms lymphocytes in immune system;
- 7 blocks lymphocyte replication;
- 8 prevents maturation of lymphocytes;
- 9 absence of infection fighting cells i.e. T and B cells;
- 10 consequences - susceptible to opportunistic infections
(pneumonia / PCP / meningitis / chicken pox / measles) and
children usually die (if not treated);

(c) Explain what is meant by gene therapy. [1]

- 1 transfer/addition of one/more normal (or modified) allele/gene into an individual's body cell;
- 2 to correct a genetic defect/a form of somatic cell therapy;
- 3 via the use of vectors (viral and non-viral) for the insertion of a transgene/to restore phenotype/normal phenotype can be expressed.

(d) Suggest why recessively inherited disorders are particularly amenable to gene therapy. [2]

- 1 such disorders are almost always simple loss-of-function mutations;
- 2 replace with one normal / dominant allele;
- 3 produce sufficient amounts of gene product;
- 4 to mask effect of recessive mutation;
- 5 for dominant disorders, both mutant alleles must be replaced/inactivated.

The ADA gene therapy approach involves essentially the following steps:

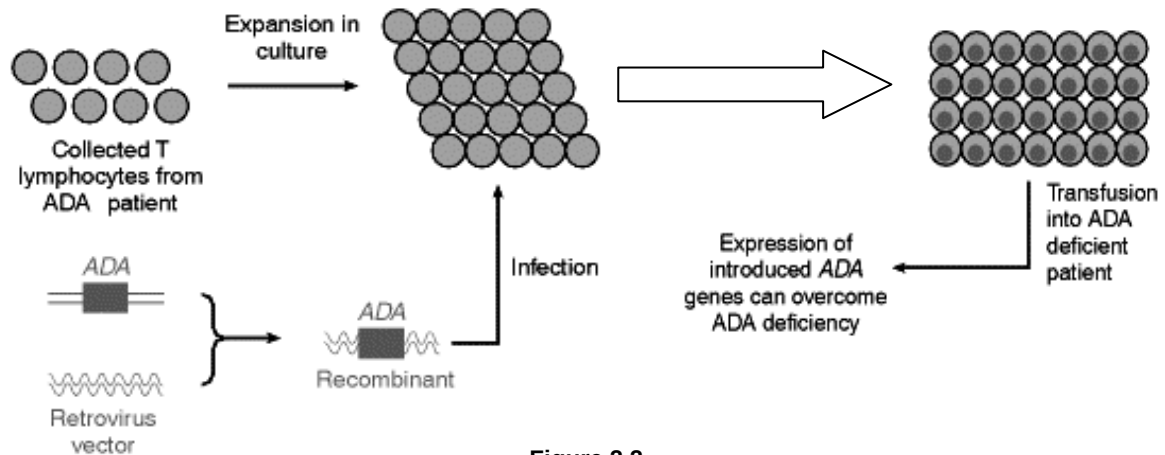


Figure 2.2

(e) Describe how gene therapy can be carried out and explain why the *ex vivo* method was used in this case. [3]

- 1 T-lymphocytes can be easily isolated;
- 2 T-lymphocytes can be cultured in the laboratory easily;
- 3 Ref use of REs to create recombinant retrovirus;
- 4 infection of cells with retrovirus containing normal ADA gene;
- 5 allows selection of / identify successfully transformed cells;
- 6 expand transformed cells through cell culture;
- 7 re-implant cells into patient;

(f) Gene therapy using the same form of treatment as the above case study succeeded for 15 French baby boys suffering from X-linked SCID. However 3 of the little boys later developed cancer and died. Suggest how this is possible. [3]

- 1 inability to control the specific site of insertion;
/ random integration of transgene;
/ insertional mutagenesis;
- 2 disrupt gene / control regions ;
- 3 activation of oncogenes/mutation of proto-oncogene ;
- 4 inactivation of tumour suppressor gene ;
- 5 inactivation of a gene controlling apoptosis;
- 6 ref. uncontrolled cell division/proliferation;

(g) Name 2 other methods, other than gene therapy, which can be used for the treatment of ADA-deficiency. [1]

- 1 **bone marrow transplantation** for a perfectly HLA-matched sibling donor
- 2 **enzyme replacement therapy** / weekly intramuscular injections of ADA

[Total : /15]

QUESTION 3

A possible route for the escape of an inserted gene from a genetically engineered crop of oil-seed rape, *Brassica napus*, is into populations of the wild turnip, *Brassica rapa*.

- (a) Explain briefly why there is a possibility of an inserted gene escaping from oil-seed rape to wild turnip. [1]

- 1 close relatives as they belong to the same genus (*Brassica*);
- 2 high possibility of cross-pollination.

Two populations of wild turnip growing next to large fields of a cultivar of oil-seed rape were studied. Seeds were collected from wild turnip plants and germinated. A small number of possible hybrids were identified by their appearance and their DNA analysed using restriction enzymes and electrophoresis.

- (b) Outline the how genetic fingerprinting is carried out for the plants. [4]

- 1 DNA is digested / cut with restriction enzymes;
- 2 at specific restriction sites;
- 3 to produce fragments;
- 4 ref gel electrophoresis;
- 5 fragments are separated by size;
- 6 DNA is loaded into the gel;
- 7 negatively charged DNA moves towards the anode;
- 8 shortest fragments travelling the furthest;
- 9 DNA is transferred by southern blotting;
- 10 use radioactive probes;
- 11 bands are visualised using autoradiography.

Some of the bands of the DNA profiles of the oil-seed rape cultivar (**A**), two wild turnip plants (**B** and **C**) and two possible inter-specific hybrids (**D** and **E**) are shown in Fig 3.1.

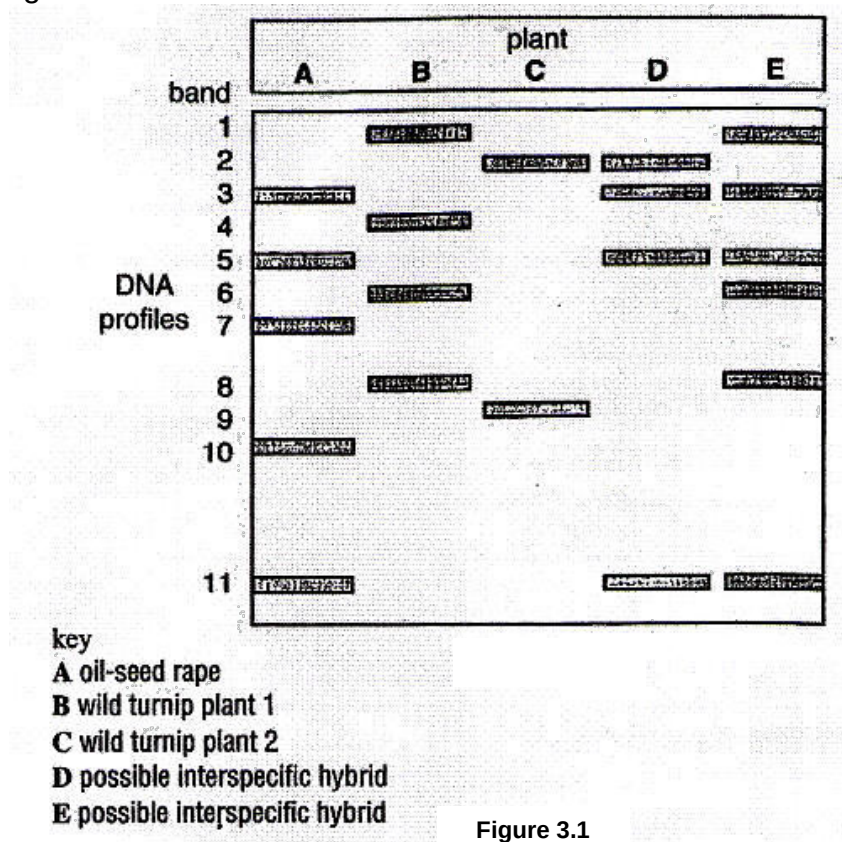


Figure 3.1

- (c) With reference to the Fig 3.1, state with reasons which plants were used to produce inter-specific hybrid plants **D** and **E**. [2]

- 1 **D – parents were A and C;**
- 2 **E – parents were A and B;**
- 3 **DNA profiles for the hybrids can be matched to some of the bands in the parents;**
- 4 **ref 3,5,11 – A, 2 – C;**
- 5 **ref 1,6,8 – B, 3, 5, 11 – A.**

- (d) Explain why there is a need to investigate hybridisation between genetically engineered crops and wild species. [2]

- 1 **risk of transfer of transgene to weeds;**
- 2 **through pollination;**
- 3 **resulting in increase invasiveness of weeds**
/ formation of super weeds;
- 4 **ref. to spread of resistance or advantageous trait**
/ herbicide resistance
/ pest resistance;

Figure 3.2 is a flow chart showing how a plasmid carrying the glyphosate-resistant gene can be used to produce glyphosate-resistant plants.

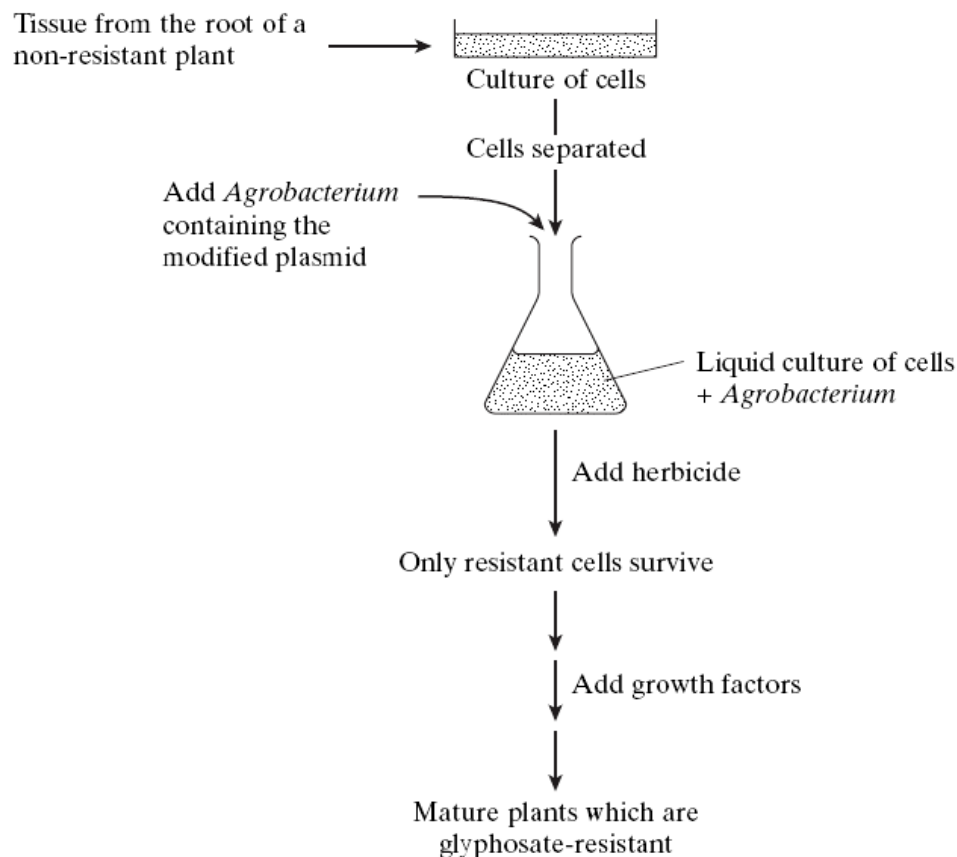


Figure 3.2

(e) With reference to Fig 3.2, describe formation of the recombinant plasmid. [3]

- 1 plasmid and the flanking region of the glyphosate resistance gene;
- 2 cut with the same restriction enzyme;
- 3 resulting in sticky ends;
- 4 ref complementary/complementary base pairs;
- 5 anneal together by H-bonds;
- 6 DNA ligase seal the nicks;
- 7 formation of phosphodiester bonds;

(f) One mature plant grows from each glyphosate-resistant plant cell. State why all cells of the mature plant contain the gene for glyphosate resistance. [1]

- 1 glyphosate resistance gene has been incorporated into the genome of the cell;
- 2 all cells are produced by mitosis / clones;
- 3 all daughter cells receive chromosome containing glyphosate-resistance gene.

(g) Explain the significance of genetic engineering in improving the quality of yield of oil seed rape. [2]

- 1** confers resistance to herbicide glyphosate;
- 2** yet able to retain their advantageous traits;
- 3** allows farmers to use herbicide to eradicate weeds;
- 4** reducing competition for nutrients from weeds;
- 5** resulting in increases in crop / quantity yield.

[Total : /15]

QUESTION 4

- (a) Explain how RFLP analysis facilitates the process of disease detection and genomic mapping. [10]

Disease detection

- 1 RFLP is based on variation in the length of DNA sequence in an allele;
- 2 The different patterns of restriction fragments is detected by the use of restriction enzymes;
- 3 used for direct / indirect screening of disease-causing alleles
- 4 ref. use before conception to screen parent's DNA
/ use after amniocentesis to screen fetal DNA;
- 5 RFLP marker that is tightly-linked (located close to disease causing allele) can serve as a surrogate / indicator for the disease-causing allele;
- 6 Crossing over between the marker and allele is unlikely to occur during meiosis, hence the marker and allele would always be inherited together;
- 7 A disease-causing allele differs from the normal allele due to mutation which alters the ability of the restriction enzyme to recognise and cut that site;
- 8 Mutation within in a restriction site within an allele may cause the restriction site to be destroyed;
- 9 resulting in the appearance of a larger fragment and the loss of the two smaller fragments;

OR

- 10 Mutation occurring in the gene may create a restriction site;
- 11 resulting in the loss of a larger fragment and the appearance of two smaller fragments;

[max 5]

Genomic mapping

- 12 RFLP markers are used to construct linkage / genetic maps of a several thousands markers evenly spaced throughout the chromosomes
- 13 RFLP markers serve as a genetic markers to order the sequence of genes;
- 14 RFLP markers can be used to determine the relative distances of genes on chromosomes;
- 15 based on the recombination frequency of two RFLP markers / an RFLP marker and a certain allele of a gene;
- 16 Recombination frequency is a measure of closeness of the two loci on the chromosome e.g. distance in cM;
- 17 example: the higher the frequency of the two loci inherited together, the closer are the two loci / vice versa;
- 18 once a genetic map is established, restriction / physical mapping determines the distance between the genetic markers (in terms of number of nucleotides (kbp) along the DNA);
- 19 each chromosome is cut by several restriction enzymes and fragments are separated by electrophoresis;
- 20 the size of the fragments produced can be estimated from the gel;
- 21 by determining the overlaps in the fragments, the original order of fragments in the chromosome and distance (kbp) can be determined;

[max 5]

- (b) An alternative method of detecting the presence of the disease allele requires the use of primers and PCR. Outline the process of Polymerase Chain Reaction. [6]

- 1 complementary primers must be synthesized/designed;
- 2 ref. 3' ends of target sequence;
- 3 ref. (automated) thermal cycler;
- 4 amplification of the target DNA sequence;

- 5 denaturation stage / ref. to 92 – 95°C;
- 6 separate double-stranded DNA into single-stranded DNA / breaking the hydrogen bonds between them;

- 7 DNA is cooled / ref. 55 – 65°C;
- 8 allow the primers to anneal to the ends of the target sequence;
- 9 by complementary base-pairing;

- 10 elongation stage / 72°C;
- 11 Taq / heat-stable polymerase synthesizes complementary strand;
- 12 by adding nucleotides to the 3' ends on both primers;

- (c) Discuss the ethical concerns of the human genome project. [4]

Discrimination and fairness

- 1 Privacy and confidentiality of genetic information;
- 2 Fairness in the use of genetic information by insurers, employers, courts, schools;
- 3 Possible discrimination for individuals with known genetic defects / prone to genetic diseases in different areas of life;
- 4 Psychological impact and stigmatization due to an individual's genetic differences;
- 5 Human responsibility and the degree of genetic influence on behaviour;

Reproductive & clinical issues

- 6 Adequacy of information for parents / individuals for decision making;
- 7 on procedures which are complex and potentially controversial;
- 8 e.g. abortion / cloning / genetic treatment of diseases / medical enhancement;
- 9 Reliability of genetic tests for pre-natal decision making;
- 10 Uncertainties associated with gene tests for susceptibilities and complex conditions linked to multiple genes and gene-environment interactions;

Ownership and commercialization

- 11 Property rights / commercialization of products (patents, copyrights and trade secrets)
- 12 Fairness in accessibility of data / material / technologies;